Pranav Venkateswaran

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EDUCATION

Thomas Jefferson High School for Science and Technology, Alexandria, VA — Expected June 2028
- *Weighted GPA:* 4.42 | *SAT:* 1490 (Aug 2026)

Rock Ridge High School, Ashburn, VA — 2024 - 2025

RESEARCH EXPERIENCE

George Mason University, Center for Infectious Disease Research, *Research Intern*, Manassas, VA — Jun 2025 - Aug 2025
- Designed antimicrobial peptides in CAMPR4, using amino acid substitutions to tune charge, hydrophobicity, and stability
- Predicted peptide structures with AlphaFold and analyzed target interactions with AutoDock Vina
- Presented findings at GMU CIDR; research published in the Mason Archival Repository Service (MARS)

NASA GeneLab for High Schools (GL4HS), *Team Lead, ArabidOMICS*, Virtual (NASA Ames) — Jul 2026 - Aug 2026
- Led a capstone team analyzing Arabidopsis spaceflight RNA-seq data (OSD-437) for spaceflight-induced expression changes
- Ran differential expression (volcano plots) and DAVID/KEGG pathway enrichment to flag spaceflight-responsive pathways
- Coordinated team deliverables across NASA's capstone modules; presented at the GL4HS Showcase (Aug 2026)

D-Select: Peptide Design for TB Breath Diagnostics, *Independent Researcher* — Summer 2026
- Built an ML pipeline that designs peptides to selectively bind TB breath biomarkers (PLAPT + DeepPurpose ensemble)
- Weighted predicted binding by breath VOC concentration to score how well each peptide separates TB from healthy breath
- Optimized by stochastic hill-climbing with ablations; lead peptide hit ROC-AUC 0.955 on held-out synthetic patient data

CLINICAL EXPERIENCE

MedStar Georgetown University Hospital, *Volunteer, S52 Med-Surg/IMC*, Washington, DC — Summer 2026
- Volunteered twice-weekly 4-hour shifts on an inpatient medical-surgical and intermediate care unit (24+ hours)

C.E.R.T. LLC, *EMT-Basic Program*, Fair Oaks, VA — Jan 2026 - Sep 2026
- Completed EMT-B course, clinical rotations, and 10/10 required patient contacts; passed Virginia psychomotor exam
- Rode along weekly with Dale City Volunteer Fire Department, observing emergency response and prehospital patient care

Washington DC VA Medical Center, *Volunteer, Diabetes Clinic*, Washington, DC — Summer 2026
- Volunteered three days a week for six weeks in an outpatient specialty clinic serving veterans with diabetes (112 hours)

Sibley Memorial Hospital (Johns Hopkins), *ED Physician Shadowing*, Washington, DC — Dec 2025 - Jul 2026
- Shadowed Dr. Nadia Eltaki in the ED (34 hours), observing patient evaluation and clinical decision-making

LEADERSHIP & SERVICE

Lincoln-Douglas Debate, *Research Coordinator*, TJHSST — 2024 - Present
- Coordinate topic research and evidence prep for the LD team; mentor newer debaters
- Varsity: Top 10, WACFL V (2025); JV: 1st Place, WACFL IV; Metrofinals Qualifier (2025)
- Logistics Coordinator for VHSL Regional debate tournaments (45+ service hours)

Loudoun County Youth Advisory Council (YAC), *Outreach Committee Member*, Loudoun County, VA — 2026 - Present
- Plan community outreach on local public health concerns for a student-led county council (Loudoun County PRCS)
- Help organize countywide service and civic engagement projects that bring teen input to local officials

American Red Cross, *Sound the Alarm Volunteer*
- Installed free smoke alarms in local homes and taught families fire safety alongside local fire departments

PROJECTS

Finance Digest, *Congressional App Challenge Submission* — 2026
- React Native app with a personalized federal tax receipt, a local federal-spending map, and a what-if budget simulator
- Built on public USAspending.gov data so users can see where their federal tax dollars go in their own community

HONORS & AWARDS

- **Technology Student Association:** State Medalist; National Qualifier (2025); Forensic Science, Video Game Design
- **Science Olympiad:** Regional Medalist; State Finalist (2024) | **Virginia JCL:** 1st Place, Latin Art (State)

SKILLS & CERTIFICATIONS

Programming: Python, Java, JavaScript, R, MATLAB, HTML/CSS, React, React Native, Node.js, PyTorch, NumPy, pandas
Bioinformatics: AlphaFold, ChimeraX, AutoDock Vina, CAMPR4, BLAST, RNA-seq differential expression, DAVID/KEGG
Lab Skills (DNA Science 1): Micropipetting, serial dilutions, Bradford protein assay, standard curves, bacterial streak plating
Certifications: AHA BLS Provider | **Languages:** English, Tamil, Latin